UNIFIED API INGESTION & ALLOCATION SPECIFICATION

Mathematical Telemetry Bus, Tensor Gate Validation, and Invariant Route Dispatching

Standard: Chi-Key Antigravity Ingestion Protocol v2.4 | **Routing Gate:** N■ × N■ ≈ N■ Invariant Check
Destination Subsystems: Cosmic Explorer (P1–4), Gravitational Resonator (P5–11), Quantum Hadron Engine (P12–21)

1. System Architecture & Ingestion Pipeline

All incoming data from external scientific telemetry, experimental detectors, AI Studio prompts, and Antigravity multi-agent subroutines enters through the **Unified Ingestion Bus**. Before any data packet is committed to the state tree or dispatched to active solver agents, it must clear the **Tensor Gatekeeper**, ensuring the mathematical parameters conform to continuous manifold emergence.

2. Ingestion Routing Rules (The 12/21 Allocation Table)

Channel / Endpoint	Payload Target & Validation	Allocated Pillar
/api/v2/telemetry/metric	g_{μν} tensor, Christoffel symbols, parametric Y(x) coordinates. Verification: det(g) ≠ 0.	Pillar 1: Intrinsic Hypersphere
/api/v2/telemetry/tensor	Ranks (N■, N■, N■). Verification rule: |(N■ × N■) - N■| / N■ < ε (Continuum check).	Pillar 2: Holographic Truth Anchor
/api/v2/telemetry/flux	Flux tube tension σ, Wilson loops, Yang-Mills field strength F_{μν}^a. Mass gap check.	Pillar 6 & 21: SQCD & Confinement
/api/v2/telemetry/cymatic	Harmonic frequencies (f = 432 Hz × 10■), nodal coordinates, soliton positions.	Pillar 20: Cymatic Particle Lattice
/api/v2/telemetry/soliton	Eguchi-Hanson instanton parameter 'a', core radius r > 0, negative energy density.	Pillar 19: Gravitational Soliton Core

3. JSON Ingestion Payload Schema

```
{ "header": { "protocol": "CHI-KEY-INGEST-2.4", "timestamp": "2026-09-08T23:45:00Z", "source_agent": "antigravity_physics_subagent_01" }, "tensor_signature": { "ranks": [64, 64, 4096], "emergence_ratio": 1.000, "melonic_trap_risk": 0.000 }, "field_state": { "throat_ratio_alpha": 0.37037, "harmonic_hz": 432.0, "vortex_winding_number": 2, "confinement_tension_gev_fm": 1.02 } }
```

Standard Ingestion Frame for Real-Time Physics Packet Reception